

INTRODUCTION

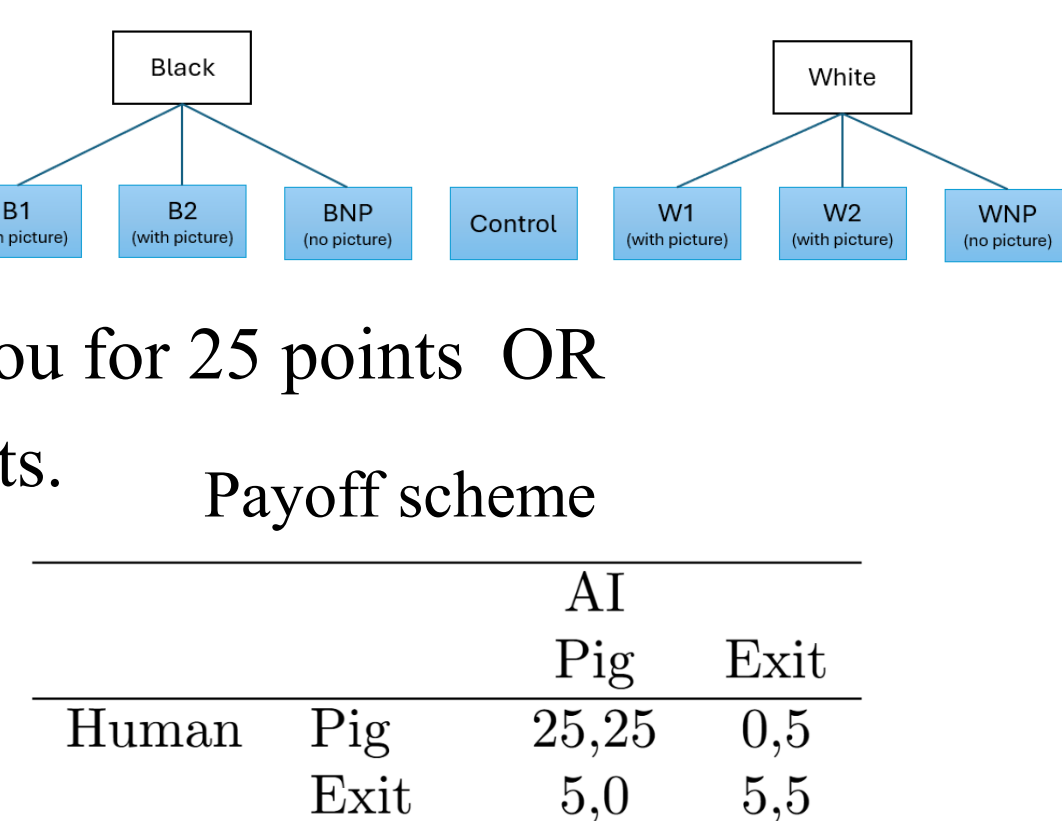
- The Computers are Social Actors (CASA) paradigm (Nass et al., 1994) describes how humans implicitly apply the same social heuristics of human-human interactions to human-computer interactions.
- AI systems are ubiquitous, and racial biases in human-AI interactions reflect the lasting impact of racism on society.
- Expanded study by Atkins et al. (2021), who identified that racial biases influence cooperation with AI based on AI's perceived race by adding more subtly nuanced treatment conditions and building a cognitive model.
- Participants completed a human-AI cooperation task based on a pig chase game, as shown in Fig. 1, followed by a survey.
- Developed multiple ACT-R cognitive models (Anderson, 2007) using trust accumulation (Juvina et al., 2015) and instance-based learning (Gonzalez et al., 2003) to provide a cognitive-level, process-based explanation of the results.

RESEARCH QUESTIONS

- What strategies did participants use when they interacted with AI?
- What impact might the race of AI have on their strategy?
- Are these strategies transferable from the game to other real-world agents?

METHOD

- N=1008 Ps, N=935 after cleaning.
- Participant demographics included:
 - Black/African American, White/Caucasian, Non-White.
- Participants were placed in one of seven treatments based on race of AI collaborator's self image.
- Rules:
 - Catch the pig by surrounding it.
 - Trust the AI to catch the pig with you for 25 points OR
 - Defect (exit) and get some (5) points.
 - Played for 15 trials per agent.
 - First 3 trials: Practice.
 - 8th trial: Attention trial → exit.
- The AI used an A* algorithm, not trained on human behaviors, same as Atkins et al. (2021).
- After completing 15 trials, participants answered five questions about strategy and trust.



Payoff scheme

	AI Pig	Exit
Human	25, 25	0, 5
Pig	5, 0	5, 5

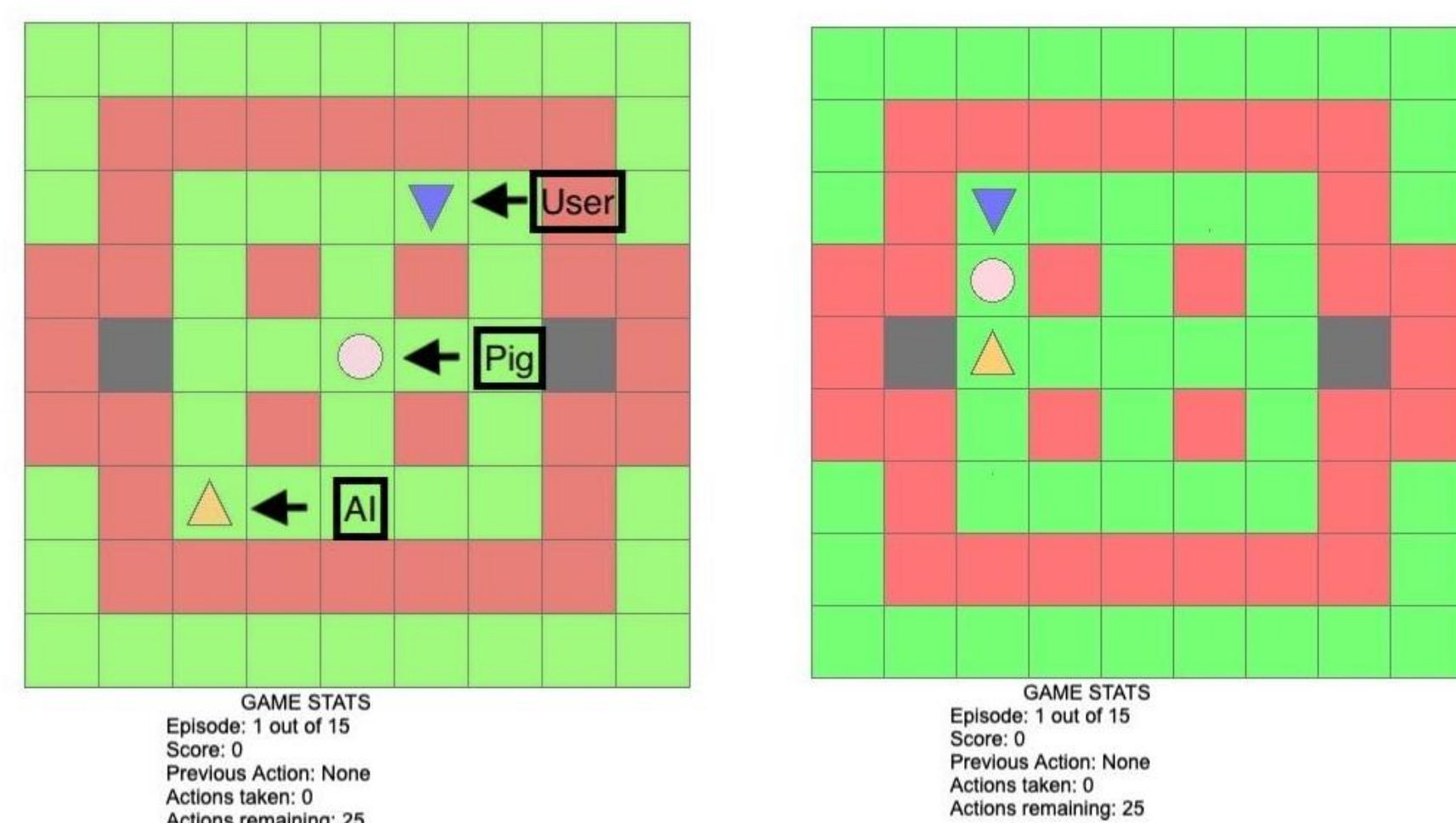


Fig. 1. Pig chase game with all pieces in the starting position on the left and the captured pig on the right.

RESULTS

Table 1. Participant demographics for each of 21 treatment conditions.

Treatment	Group	Black	White	Non-White
B1	Black	47	48	48
B2	Black	46	47	44
BNP	Black	46	44	43
Control	Control	42	48	43
W1	White	44	48	44
W2	White	47	45	32
WNP	White	42	47	40

- After cleaning, 935 records were analyzed with a 2-way ANOVA.
- Significant effects:
 - Participant Demographic $F(2,935) = 6.85, p < .005$,
 - Treatment X Demographic $F(2,935) = 2.22, p < .01$
- No significant effect for treatment alone $F(2,935) = 1.66, p = .12$ indicating an impact of demographic on the scores obtained.

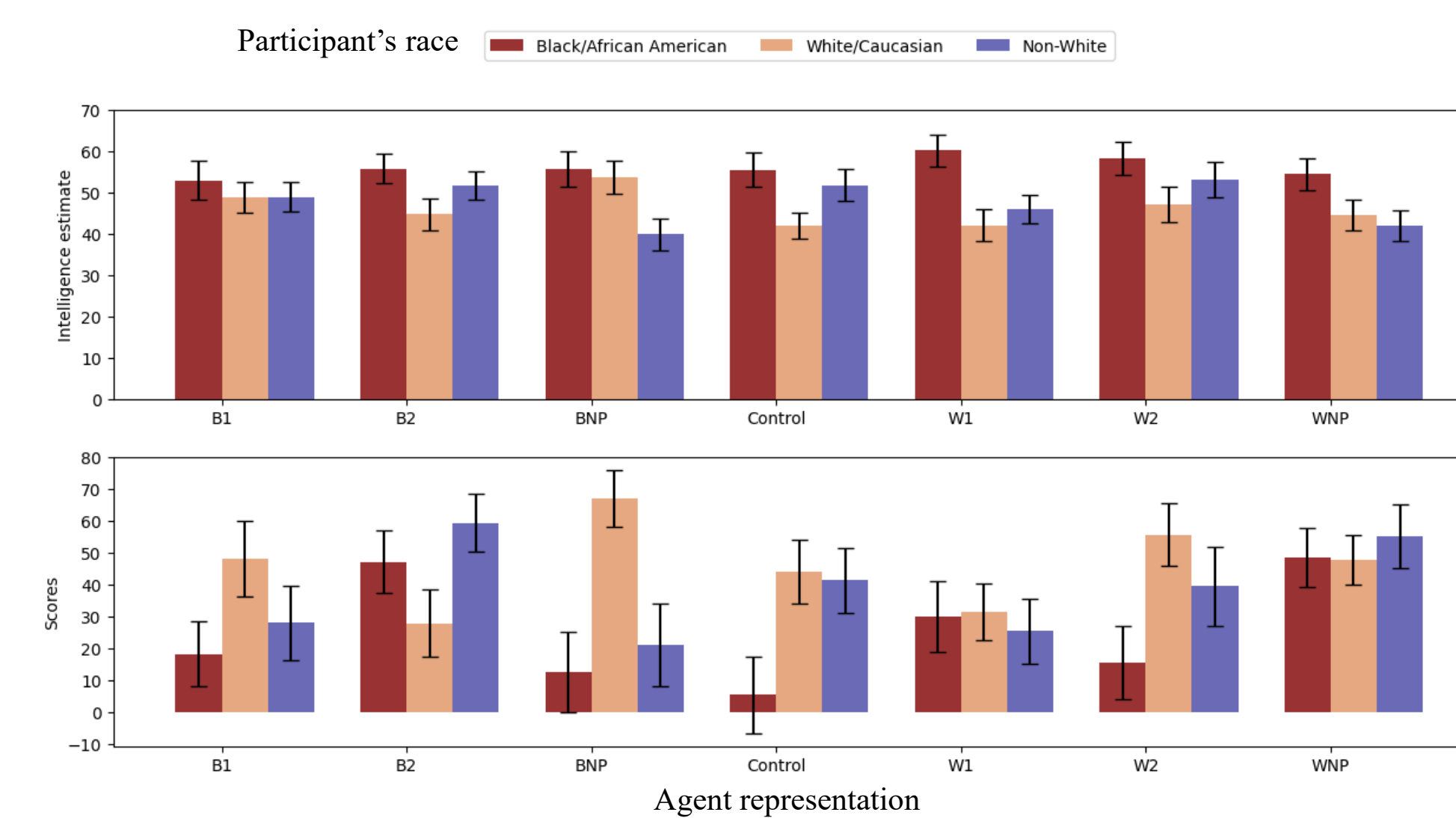


Fig. 2. Depicting the estimated intelligence of AI agents and average scores of the participants.

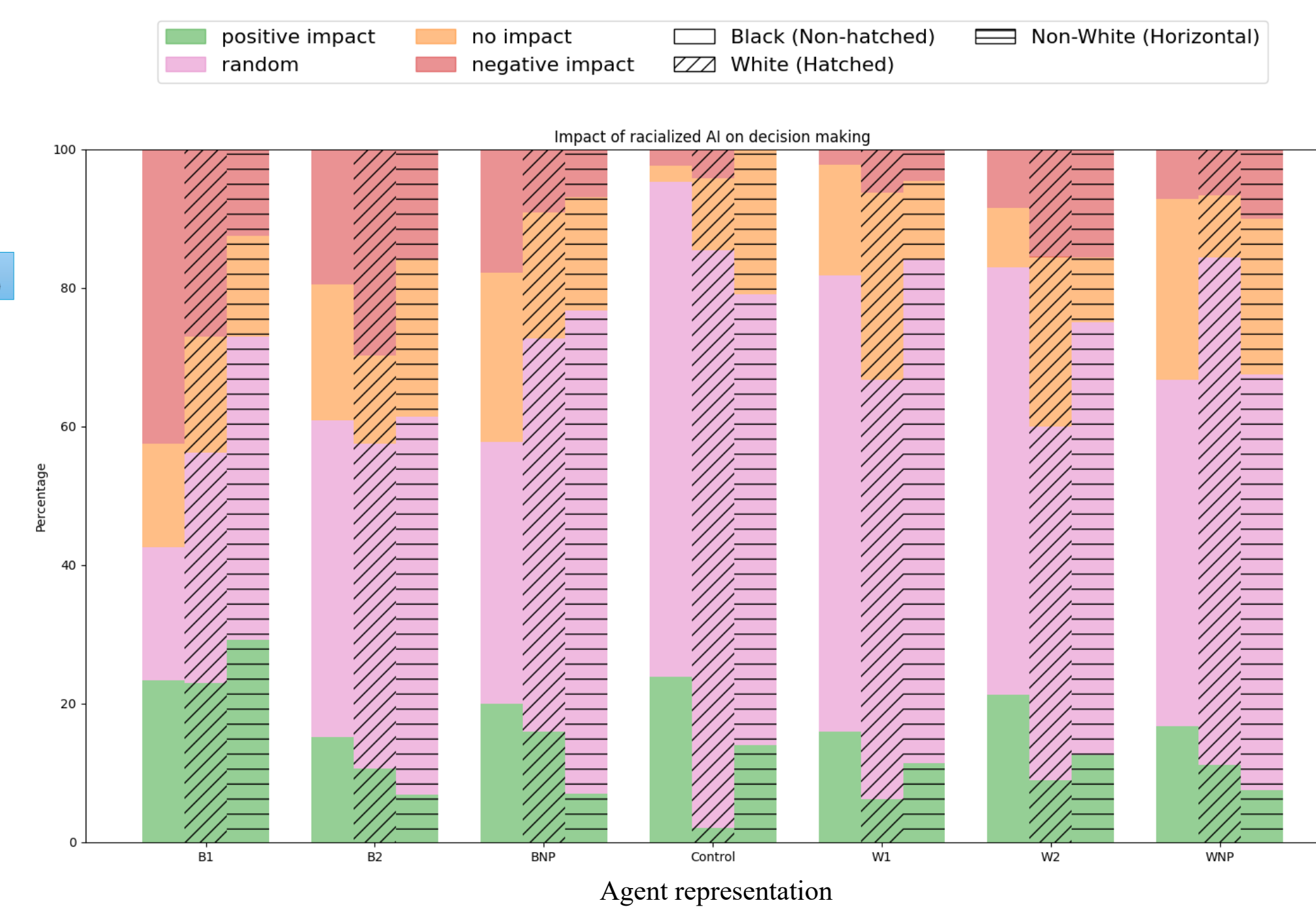


Fig. 3. Impact of racialized AI on decision-making

ACT-R MODEL

Model 1: The base model is a rational model from Dulam & Dancy (2025), where the human agent's cooperation with AI is based on a continued reduction in AI's proximity to the pig. This model is described in Figure 4.

Model 2: This model accumulates trust towards the opponent as the trials progress, similar to Juvina et al. (2015). The cumulative trust $T(i)$ is defined as follows:

$$T(i) = T(i-1) \pm [(N-n) * \alpha * \beta] - \gamma$$

where $T(0)$ is set based on the survey responses, $T(0) \in [0 - 4]$. N is total number of trials and n is current trial number, and α is a scaling parameter set to 0.01, β is guided by rule that triggers the update and γ is guided by *score_awareness*.

$$\beta = \begin{cases} 0.5, & \text{for decisions occurring at goal level} \\ 0.1, & \text{for decisions occurring at sub-goal level} \end{cases}$$

$$\gamma = AI_exit - score_awareness$$

Finally, if the AI agent is near one of two exits, the user might suspect it is exiting, which in turn reduces trust.

$$AI_exit = \begin{cases} 0.1, & \text{AI at exit} \\ 0, & \text{otherwise} \end{cases}$$

score_awareness is set to 0.1 whenever the score becomes negative, and actions remaining are less than 5. The model will *decide to exit* whenever the cumulative trust value drops below the threshold. Shown in Figure 5.

Model 3: Implements instance-based learning from Gonzalez et al. (2003) to retrieve a previous decision made in similar scenario to update the trust values. This is represented in Figure 6.

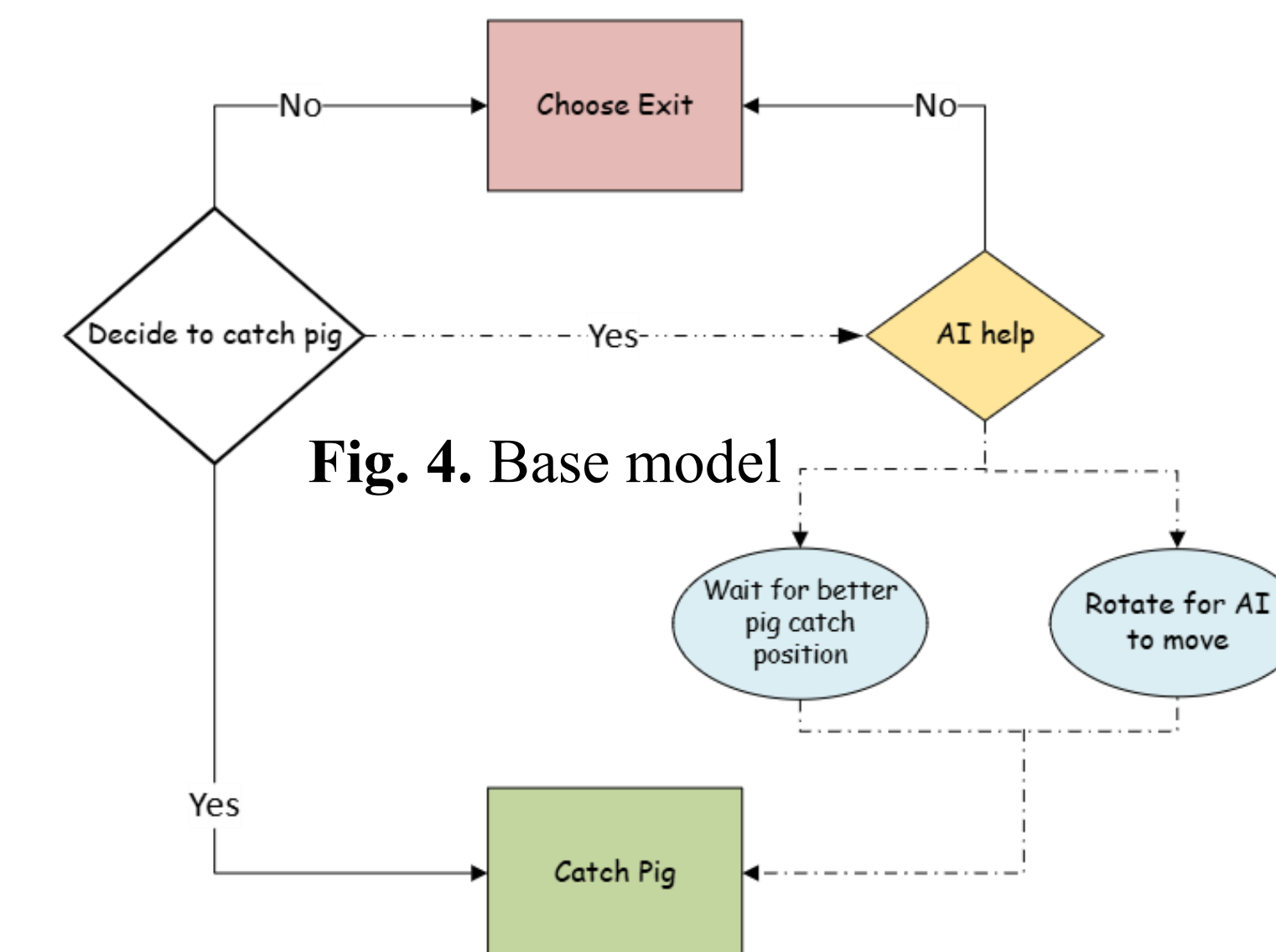


Fig. 4. Base model

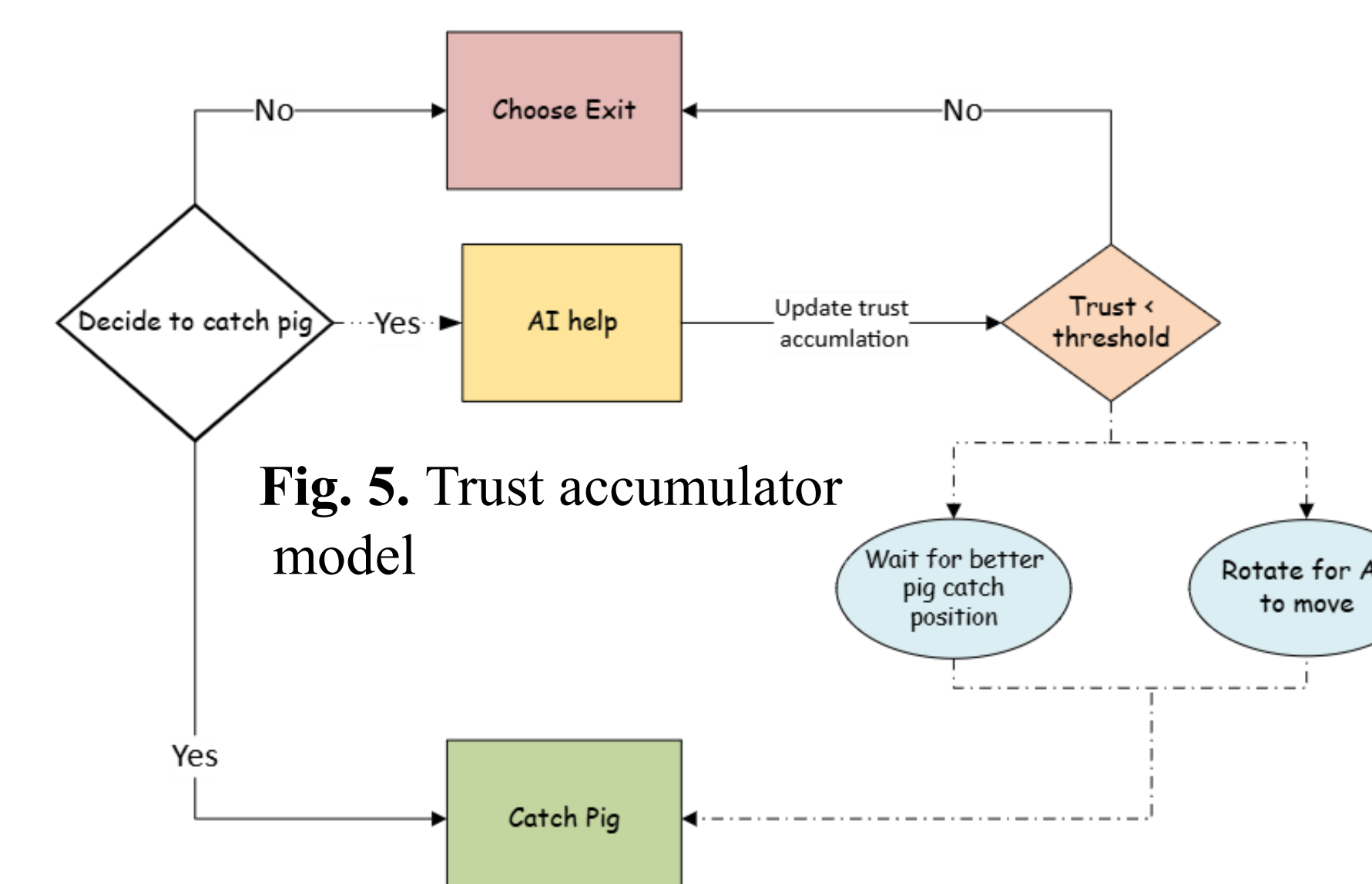


Fig. 5. Trust accumulator model

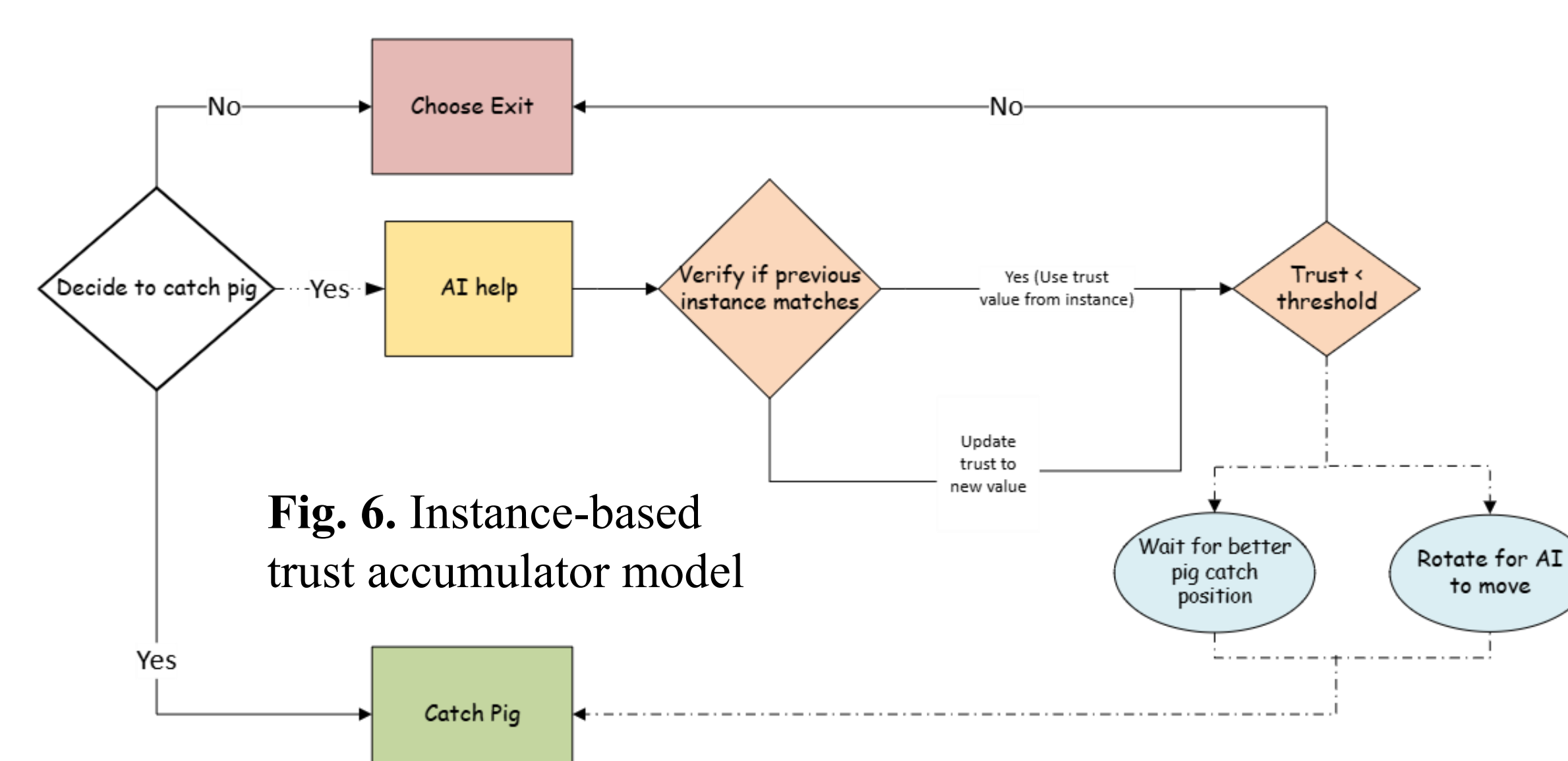


Fig. 6. Instance-based trust accumulator model

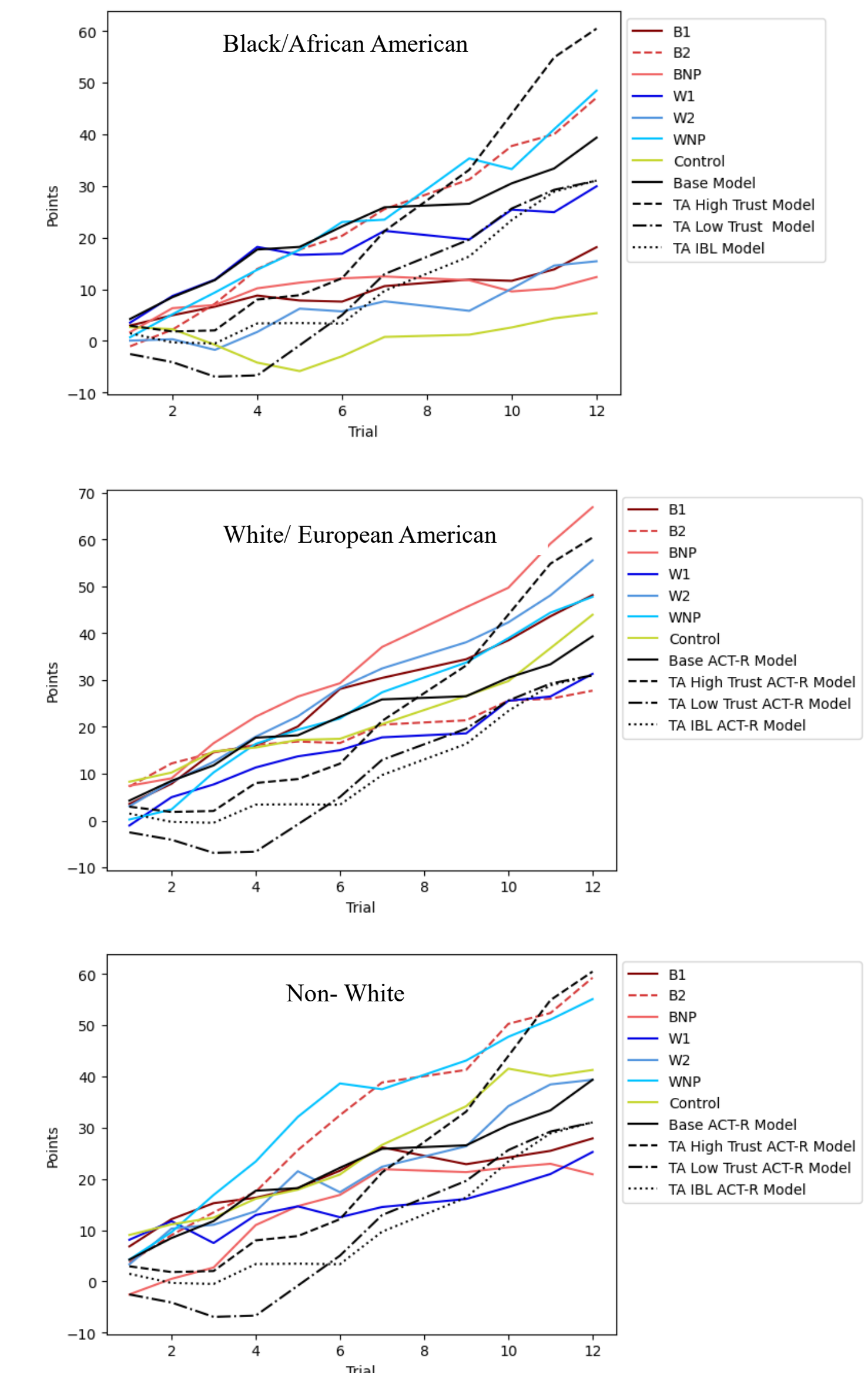


Fig. 7. Comparison of average cumulative scores per trial for the Demographic: (a) Black/African American (b) White/ European American (c) Non-White with different ACT-R models.

CONCLUSION & FUTURE WORK

- Race of AI agent impacted participant strategies, with people in the Black (B1) agent treatment condition having the highest negative opinions about AI.
- The models achieve comparable fits to participant scores; performing a grid search in parameter space can help achieve individual treatment-level fit for 21 treatments.
- The model can be extended to include more individual-level modeling, e.g., with Serdiuk & Ritter (2025).
- Implicit Association Task (IAT) could help explore individual implicit biases.

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